

MOTHLINE

3-2-1 BACKUP & RECOVERY GUIDE

CachyOS + Niri + Noctalia v5

The objective

Maintain three copies of the Mothline configuration, on two different kinds of storage, with one encrypted copy kept off-site. The guide is tailored to Zsh and the current Mothline configuration paths.

1 // LAPTOP	2 // EXTERNAL DRIVE	3 // ENCRYPTED CLOUD
Fast local recovery	Offline, physically separate	Survives theft or local damage

Recommended baseline: keep the plain archive on the laptop for convenience, and keep only encrypted copies outside the laptop. Store the encryption passphrase in Bitwarden and never inside the backup folder.

1. CREATE THE MOTHLINE RESTORE POINT

The archive below captures the active desktop-rice configuration plus an explicit package list. It intentionally includes Claws Mail appearance settings but not the entire mail directory.

```
mkdir -p ~/Mothline-Backup

pacman -Qqe > ~/Mothline-Backup/explicit-packages.txt

tar -czf ~/Mothline-Backup/mothline-config-$(date +%Y%m%d-%H%M%S).tar.gz \
-C ~ \
.config/noctalia \
.config/niri \
.config/gtk-3.0 \
.config/gtk-4.0 \
.config/qt6ct \
.config/kitty \
.config/MangoHud \
.config/starship.toml \
.claws-mail/clawsrc
```

Create and verify a checksum:

```
sha256sum ~/Mothline-Backup/*.tar.gz > ~/Mothline-Backup/SHA256SUMS

cd ~/Mothline-Backup
sha256sum -c SHA256SUMS
```

Optional additions

If you later want a broader rice snapshot, add the specific Firefox chrome/profile customization files, Steam skin files, wallpapers, avatars, SDDM theme directory, and start-page project. Avoid archiving complete browser profiles because they may contain cookies, sessions, and other sensitive data.

Inventory the result:

```
ls -lh ~/Mothline-Backup
tar -tzf ~/Mothline-Backup/mothline-config-YYYYMMDD-HHMMSS.tar.gz | less
```

2. ENCRYPT BEFORE THE BACKUP LEAVES THE LAPTOP

The archive may reveal usernames, paths, application settings, mail-server metadata, and installed packages. Encrypt it before placing it on removable media or cloud storage.

Install age

```
sudo pacman -S age
```

Passphrase encryption

```
age -p \  
-o ~/Mothline-Backup/mothline-config-YYYYMMDD-HHMMSS.tar.gz.age \  
~/Mothline-Backup/mothline-config-YYYYMMDD-HHMMSS.tar.gz
```

Use the exact archive name shown by `ls ~/Mothline-Backup/*.tar.gz`. Save the passphrase in Bitwarden. Do not save it in a text file beside the archive.

Test decryption before trusting the copy

```
age -d \  
-o /tmp/mothline-backup-test.tar.gz \  
~/Mothline-Backup/mothline-config-YYYYMMDD-HHMMSS.tar.gz.age  
  
tar -tzf /tmp/mothline-backup-test.tar.gz | head  
rm /tmp/mothline-backup-test.tar.gz
```

Stronger long-term option

For unattended or scripted backups, use an age identity file instead of a typed passphrase. Keep the private identity offline and in a password manager attachment. Never upload the private identity with the archive.

Checksum the encrypted file

```
cd ~/Mothline-Backup  
sha256sum *.age > SHA256SUMS-ENCRYPTED  
sha256sum -c SHA256SUMS-ENCRYPTED
```

3. COPY 2: EXTERNAL USB DRIVE OR SSD

A small USB flash drive is adequate for this configuration archive. An external SSD is preferable if you later include wallpapers, game configurations, or larger personal data.

Identify the mounted drive

```
lsblk -f
findmnt --real
```

The mount point will often resemble `/run/media/skilgore/DRIVE-NAME`. Confirm the exact path before copying.

Copy and verify

```
cp -av ~/Mothline-Backup /run/media/skilgore/DRIVE-NAME/

cd /run/media/skilgore/DRIVE-NAME/Mothline-Backup
sha256sum -c SHA256SUMS-ENCRYPTED
```

Safely unmount

```
sync
udisksctl unmount -b /dev/sdX1
```

Physical-storage practice

Label the drive 'Mothline Backup', add the backup date, disconnect it after verification, and keep it somewhere separate from the laptop. A permanently connected drive is vulnerable to the same accidental deletion, malware, electrical damage, or theft.

Useful applications

Tool	Purpose	Install
age	Simple modern encryption	<code>sudo pacman -S age</code>
Pika Backup	Friendly Borg-based desktop backups	<code>flatpak install flathub org.gnome.World.PikaBackup</code>
restic	Encrypted, deduplicated, scriptable backups	<code>sudo pacman -S restic</code>
rclone	Copy encrypted files to cloud providers	<code>sudo pacman -S rclone</code>

4. COPY 3: ENCRYPTED OFF-SITE CLOUD COPY

Google Drive, Dropbox, or OneDrive are all suitable for the encrypted archive. Upload only the .age file, the package list, and the encrypted-file checksum. Do not upload the unencrypted .tar.gz archive.

Simple manual upload

Open your chosen provider, create a folder named **Mothline Backup**, and upload:

```
mothline-config-YYYYMMDD-HHMMSS.tar.gz.age
explicit-packages.txt
SHA256SUMS-ENCRYPTED
```

Optional rclone workflow

```
sudo pacman -S rclone
rclone config

# After creating a remote named cloud:
rclone copy ~/Mothline-Backup cloud:Mothline-Backup \
--include '*.age' \
--include 'explicit-packages.txt' \
--include 'SHA256SUMS-ENCRYPTED' \
--progress
```

Cloud security

Enable multi-factor authentication on the cloud account. Keep recovery codes offline. Encryption protects the archive contents, while MFA protects access to the account and backup history.

Verify a downloaded cloud copy

```
mkdir -p ~/Mothline-Backup-Verify
rclone copy cloud:Mothline-Backup ~/Mothline-Backup-Verify --progress
cd ~/Mothline-Backup-Verify
sha256sum -c SHA256SUMS-ENCRYPTED
```

Retention recommendation

Keep the three newest dated archives and one known-good milestone archive. This provides rollback protection without creating an unmanageable collection. Never overwrite the only known-good copy.

5. RESTORE AND RECOVERY

Restore carefully

Do not extract blindly over a live configuration. Decrypt into a temporary directory, inspect the archive, stop the affected applications, and restore only the paths you need.

Decrypt and inspect

```
mkdir -p ~/Mothline-Restore
age -d \
-o ~/Mothline-Restore/mothline-config.tar.gz \
/path/to/mothline-config-YYYYMMDD-HHMMSS.tar.gz.age
tar -tzf ~/Mothline-Restore/mothline-config.tar.gz | less
```

Extract into a staging folder

```
mkdir -p ~/Mothline-Restore/staging
tar -xzf ~/Mothline-Restore/mothline-config.tar.gz \
-C ~/Mothline-Restore/staging
```

Compare before replacing

```
diff -ruN ~/.config/noctalia \
~/Mothline-Restore/staging/.config/noctalia | less
```

Restore one component

```
cp -a ~/.config/noctalia ~/.config/noctalia.before-restore
cp -a ~/Mothline-Restore/staging/.config/noctalia ~/.config/
noctalia msg config-reload
```

Reinstall explicitly selected packages on CachyOS/Arch

```
sudo pacman -S --needed - < explicit-packages.txt
```

AUR packages may require a separate review. List packages not available in configured repositories with:

```
comm -23 <(sort explicit-packages.txt) <(pacman -Slq | sort -u)
```

After restoration, log out and back in if Niri environment variables or Qt platform-theme settings changed.

6. MAINTENANCE CHECKLIST

When	Action
After a major Mothline change	Create a new dated archive; encrypt; verify; copy to USB and cloud.
Monthly	Run both checksum files and test-list the newest encrypted archive.
Every 3 months	Perform a staging-folder restore test; confirm Noctalia, GTK, Qt, and Claws paths exist.
Before an OS reinstall	Create a fresh archive, export the package list, and verify both off-device copies.
After replacing hardware	Restore only after confirming ownership, paths, monitor names, and package versions.

Quick health check

```
cd ~/Mothline-Backup
sha256sum -c SHA256SUMS
sha256sum -c SHA256SUMS-ENCRYPTED
age -d -o /tmp/mothline-check.tar.gz newest-backup.tar.gz.age
tar -tzf /tmp/mothline-check.tar.gz >/dev/null && echo 'Backup test: OK'
rm /tmp/mothline-check.tar.gz
```

Best practical setup for Mothline

Laptop copy for fast recovery + disconnected USB drive at home + encrypted Google Drive copy. Keep the passphrase in Bitwarden, verify monthly, and retain three recent versions plus one milestone version.

Reference links

age: <https://age-encryption.org/> | restic: <https://restic.net/> | rclone: <https://rclone.org/> | Pika Backup: <https://apps.gnome.org/PikaBackup/>